Introduction to Information Security

Firewalls

What is a Firewall?

- A **point** of control and monitoring
- Imposes restrictions on network services
  - only authorized traffic is allowed
- Controlling access
  - can implement alarms for abnormal behavior
- A firewall is a security policy enforcement point that regulates access between computer networks
- Controls TCP protocols
  - http, smtp, ftp, telnet etc

What do Firewalls Protect?

- Data
  - Financial information
  - Sensitive employee or customer data
- Resources
  - Computing resources
  - Time resources
- Reputation
  - Loss of confidence in an organization
  - Intruder uses an organization's network to attack other sites

Who do Firewalls Guard Against?

- Internal Users
- Hackers
  - block unauthorized access to networks and systems by filtering incoming and outgoing traffic,
- Corporate Espionage
  - obtain confidential business information to gain a competitive advantage
- Terrorists
- Common Thieves
  - prevent malicious individuals from exploiting vulnerabilities

Common Internet Threats

- Denial of service attacks
  - Specific attacks that can cause a server crash
  - Flooding the server with traffic to disrupt or deny service
- Intrusion threats
- Attacks on services
  - The backend server may not be enough for adequate protection, but the firewall can block external attacks

How Vulnerable are Internet Services?

- E-mail or smtp – Simple Mail Transfer Protocol
  - Risks Include
    - E-mail bombing (stalking)
      - Anonymous harassment
      - Large amounts of e-mail to a single user address
    - Spamming
      - Messages sent to numerous different users from a host
    - Virus download mechanism

How Vulnerable are Internet Services?

- FTP - File Transfer Protocol
  - Risks Include
    - Unencrypted authentication
      - Usernames and passwords can be "sniffed"
    - Unencrypted data transfers
      - Data can be viewed

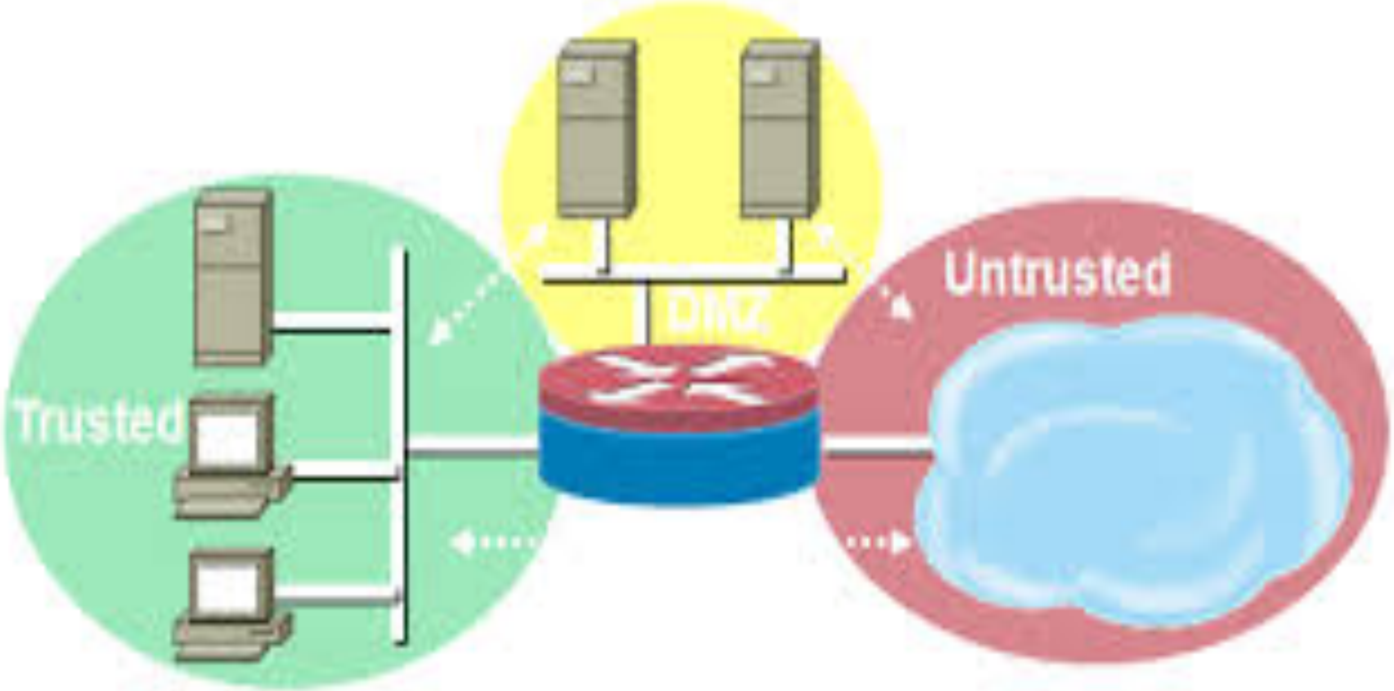
How Vulnerable are internet services?

- HTTP – Hypertext Transfer Protocol
  - Risks Include
    - Browsers can be used to run dangerous commands
    - Difficult to secure
    - Remote execution of commands and execution (server side)

How Vulnerable are Internet Services?

- DNS (Domain Name System)
  - Risks include
    - DNS cache poisoning
      - If an attacker can inject malicious records into the cache, they can direct users to fraudulent websites.
    - DNS spoofing
      - Attacker intercepts DNS queries sent by the victim and returns malicious DNS responses, redirecting them to fake websites or to attacker-controlled servers.

So what does a Firewall do?



Firewall Architecture Overview

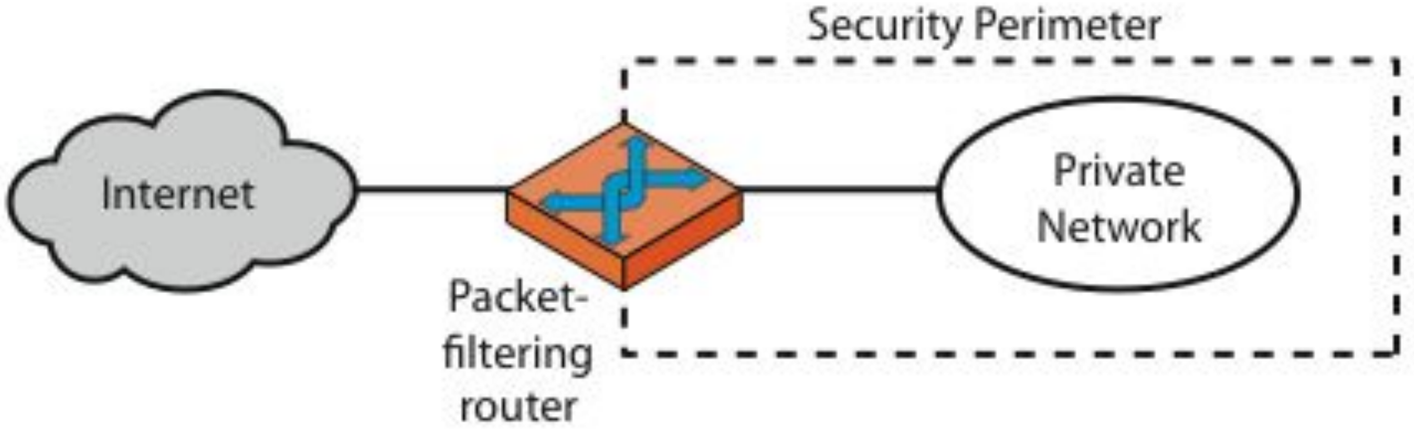
- Basic Firewall Components
  - Software
  - Hardware
  - Purpose Built/Appliance based

Classification of Firewall

Characterized by protocol level it controls in

- Packet filtering
- Circuit gateways
- Application gateways
- Combination of above is dynamic packet filter

Firewalls – Packet Filters



(a) Packet-filtering router

Firewalls – Packet Filters

- Simplest of components
- Uses transport-layer information only
  - IP Source Address, Destination Address
  - Protocol/Next Header (TCP, UDP, ICMP, etc)
  - TCP or UDP source & destination ports
  - TCP Flags (SYN, ACK, FIN, RST, PSH, etc)
  - ICMP message type

Usage of Packet Filters

- Filtering with incoming or outgoing interfaces
  - E.g., Ingress filtering of spoofed IP addresses
  - Egress filtering

Ingress Filtering

- Filter malicious traffic coming from outer untrusted network (Internet).
- Checking the source IP Addresses of incoming packets to prevent malicious traffic
- Block packets in case if Source IP Address is:
  - IP Address of internal network
  - Loopback IP Addresses
    - IP addresses used by a device to send network traffic to itself. The device can communicate with itself as if it were communicating with another machine on the network
  - Multicast IP addresses
    - Allows the sender to send one copy of the data to a group of recipients
  - Private IP Addresses from outer network

Egress Filtering

- Monitoring and controlling the outbound malicious traffic from internal network to outer network.
- Monitoring the Source IP Addresses of packets going from internal network to outer network.
- If the source IP address is not in the trusted IP Address list of internal network then IP Spoofing attack is launched by some one from your internal network.

Firewall Gateways

- A **firewall gateway** is a security device or system that acts as a barrier between two or more networks, typically a private network
  - Filter incoming, outgoing packets
  - All incoming traffic directed to firewall
  - All outgoing traffic appears to come from firewall
  - Operates at the **Transport Layer (Layer 4)** of the OSI model

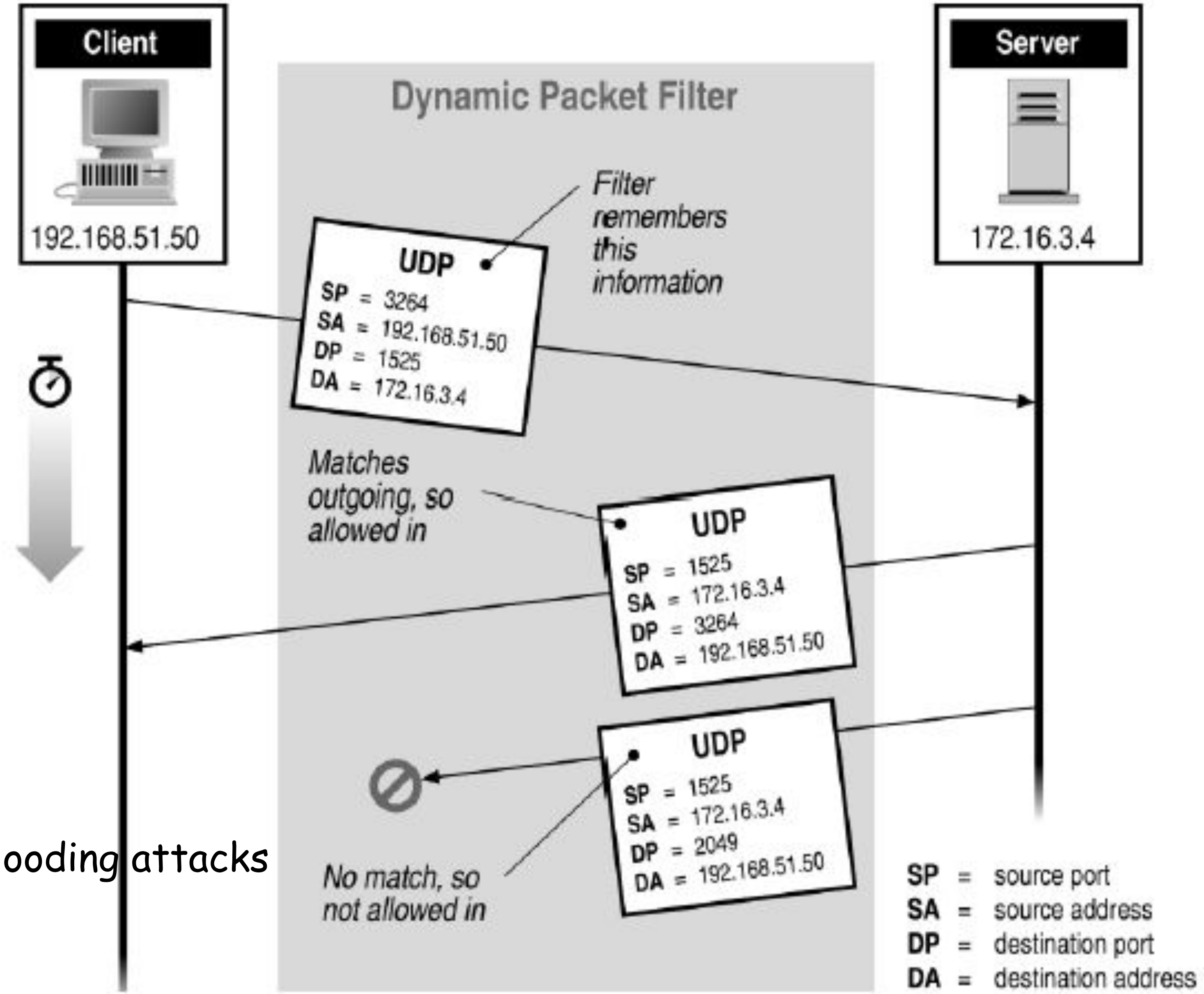
Application-Level Filtering

- Acts as an intermediary that filters traffic based on **application-specific protocols** (e.g., HTTP, FTP, SMTP, DNS).
- It inspects the actual content of the data being transferred, ensuring that it matches the expected application behavior.
- Operates at the **Application Layer** of the OSI model

Dynamic Packet Filter Implementation

- Dynamically update packet filter's ruleset
  - Rules from the rule set can be selected on conditional basis.
  - Uses State Table for getting information about connections between hosts.
  - Rules can be selected at run time.
  - Functionality of different layers of firewall can also be selected dynamically depending upon nature of traffic.

Stateful Filtering



To avoid TCP SYN flooding attacks